

# Computer Networks

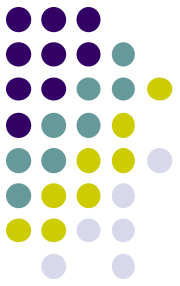
## EE357

### Introduction-Part 1

Haiming Jin

The slides are adapted from those provided by Prof. Shizhen Zhao.

# Chapter 1: Introduction

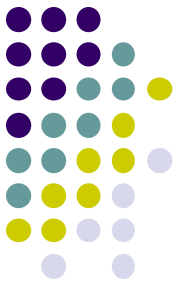


## Our goal:

- get “feel” and terminology
- paint a broad picture
- see the forest through the trees
- approach:
  - use Internet as example

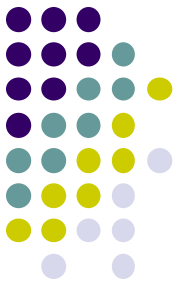
## Topics:

- What’s Computer Network?
- Protocol layers, service models
- What’s the Internet?
- Network edge
- Access net and physical media
- Network core
- Internet structure and ISPs
- Delay, loss, and throughput in packet-switched networks
- History of Internet



# Chapter 1: roadmap

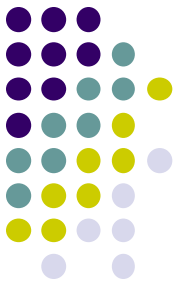
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# What is Computer Network

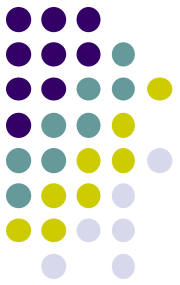
- Collection of autonomous computers interconnected by **a single technology**
  - From Computer Network by Tanenbaum
- A collection of computers and devices interconnected by communications channels that facilitate communications among users and allows users to share resources.
  - From wikipedia

# Classification of network by physical media



- bit propagates between transmitter/receiver pairs in the form of electromagnetic waves or optical pulse across physical media.
- **physical media:** what lies between transmitter & receiver
  - **guided media:** signals propagate in solid media: twisted pair, fiber optics, coaxial cable
  - **unguided media:** signals propagate freely, e.g., atmosphere, outer space

# Guided Media: Twisted Pair



- Two insulated copper wires, twisted like a DNA string (reduces electrical interference). Often twisted pairs go by the bundle.
- Due to their adequate performance and low cost, twisted pairs are widely used.
- UTP (Unshielded Twisted Pair)
  - CAT3 8 cores, 16MHz bandwidth, 10Mbps
  - CAT5 8 cores, 100MHz bandwidth, 100Mbps



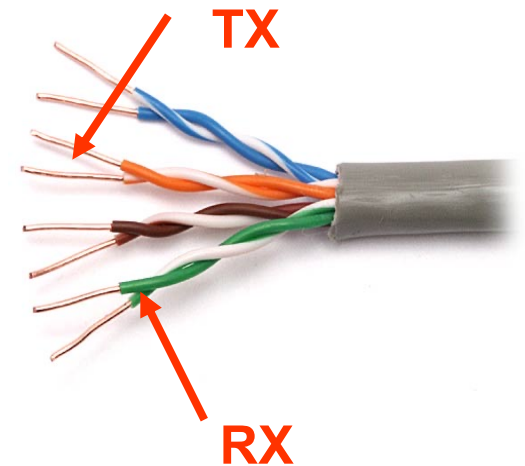
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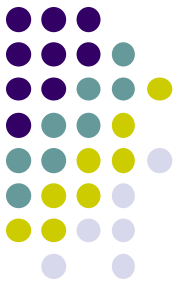
(a) Category 3 UTP



(b)

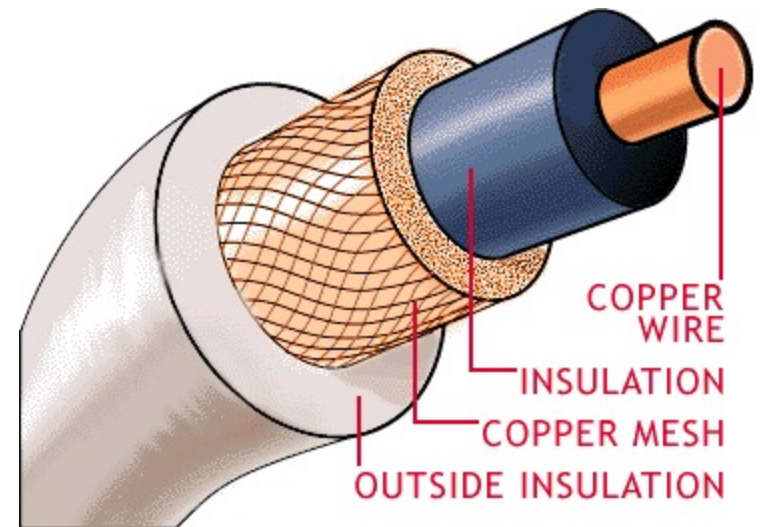
(b) Category 5 UTP.





# Guided Media: coaxial cable

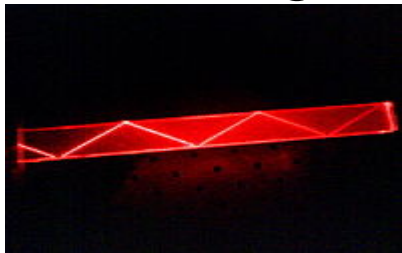
- The construction and shielding of the coaxial cable give it a good combination of high bandwidth and excellent noise immunity.  
1GHz
- Coax is better than twisted pair when you need more bandwidth, but is now rapidly being replaced with fiber.
- baseband:
  - single channel on cable
  - legacy Ethernet
- broadband:
  - multiple channels on cable
  - HFC



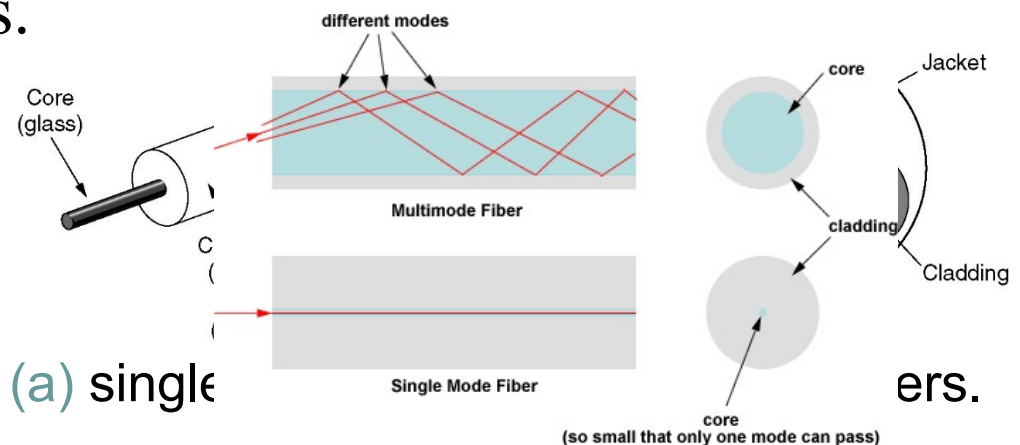
# Guided Media: Fiber Optics



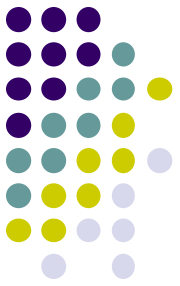
- glass fiber carrying light pulses, each pulse is a bit, which is surrounded by a glass cladding with a lower index of refraction than the core, to keep all the light in the core.
- ultra wide bandwidth: 10's THz, 10's-100's Gps
- low error rate: immune to electromagnetic noise
- ultra low attenuation : repeaters spaced far apart, can be used for long distances.



Total internal reflection



# Unguided media: radio



- signal carried in electromagnetic spectrum
- no physical “wire”
- propagation environment effects:
  - reflection
  - obstruction by objects
  - interference

## Radio link types:

- ❑ **terrestrial microwave**
  - ❖ e.g. up to 45 Mbps channels
- ❑ **LAN** (e.g., Wifi)
  - ❖ 11Mbps, 54 Mbps
- ❑ **wide-area** (e.g., cellular)
  - ❖ 4G cellular: ~ 100 Mbps
- ❑ **satellite**
  - ❖ Kbps to 45Mbps channel (or multiple smaller channels)
  - ❖ 270 msec end-end delay
  - ❖ geosynchronous versus low altitude

# Classification of network by physical media



- Wired network:
  - Twisted-pair copper wire
  - Coaxial cable
  - Optical fiber
- Wireless network:
  - Wi-Fi
  - Cellular System
  - Satellite

# Classification of network by Topology

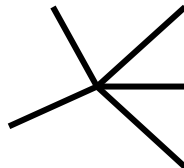


- **Network topology** is the coordination by which devices in the network are arranged in their logical relations to one another, independent of physical arrangement.

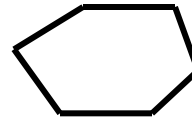
**Bus**



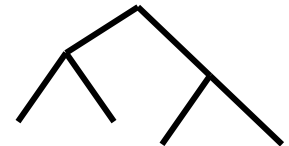
**Star**



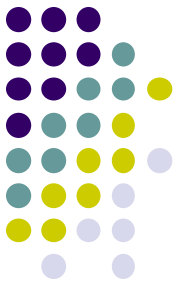
**Ring**



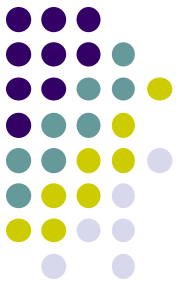
**Tree**



# Classification of network by scale

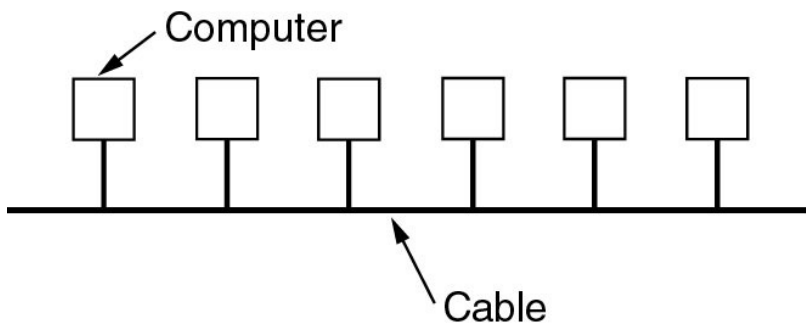


| Interprocessor distance | Processors located in same | Example                         |
|-------------------------|----------------------------|---------------------------------|
| 1 m                     | Square meter               | Personal area network (PAN)     |
| 10 m                    | Room                       | Local area network (LAN)        |
| 100 m                   | Building                   |                                 |
| 1 km                    | Campus                     |                                 |
| 10 km                   | City                       | Metropolitan area network (MAN) |
| 100 km                  | Country                    | Wide area network (WAN)         |
| 1000 km                 | Continent                  |                                 |
| 10,000 km               | Planet                     | The Internet                    |

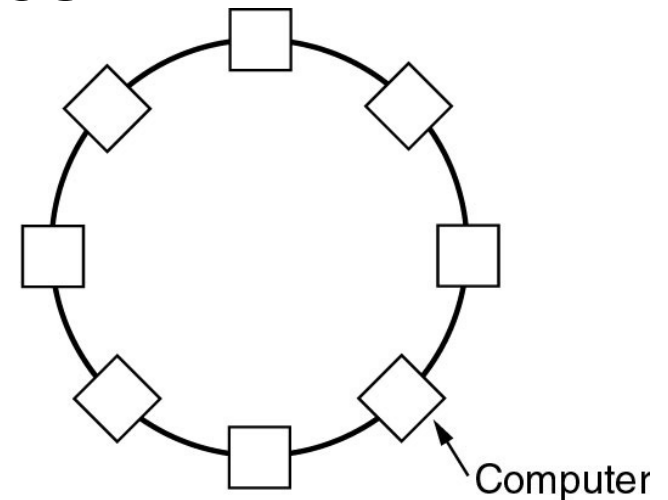


# Local Area Networks

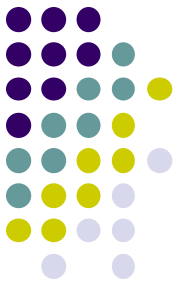
- Apart from scale, LANs distinguish themselves from other networks by (generally) using **broadcast** technology,
- and having simple **topologies**:



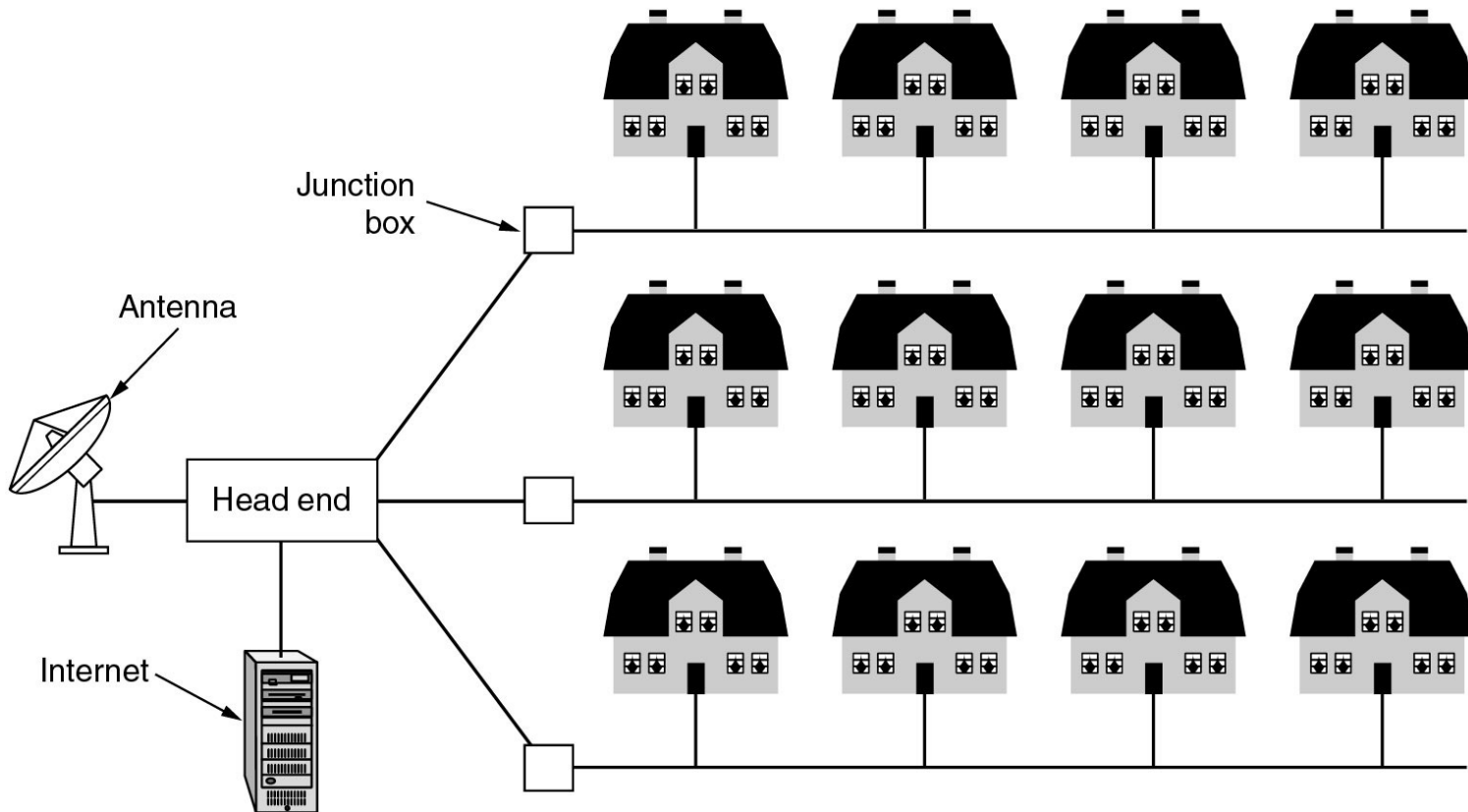
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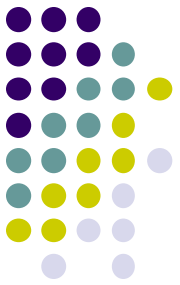
(b)



# Metropolitan Area Networks

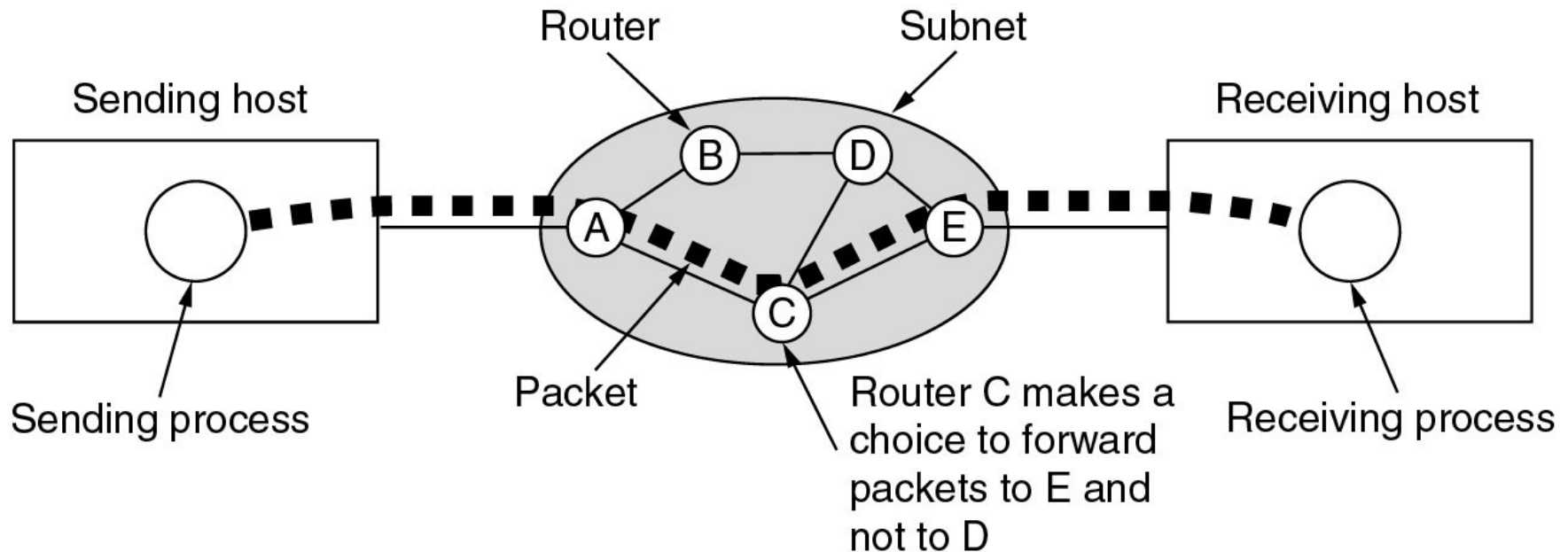


A metropolitan area network based on cable TV.

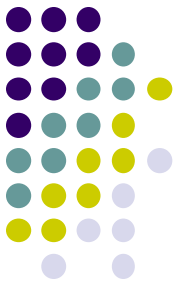


# Wide Area Networks

- A stream of packets from sender to receiver.



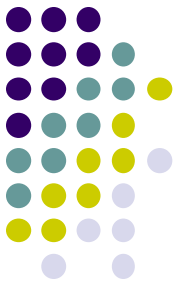
**Communication Subnet**  
**Network Core / backbone**



# Internetworks

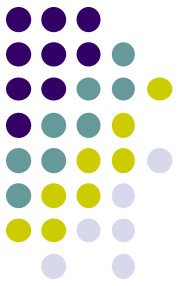
- The assumption so far is that a network is **homogeneous**: there is hardly any variation in hardware and software. In practice, large networks can only be constructed by **interconnecting** different kinds of **heterogeneous networks**=> internet(work).
- **internet(work)**: connecting LANs to each other through a WAN
- Connecting WANs to each other (e.g., the **Internet**).

# Components of Computer Networks



- **Hardware:** how you can connect computers into a network:
  - Network interface cards / Network Adapter
  - Repeaters
  - Bridges
  - Switches
  - Routers
  - Firewalls
- **Software:** This is what actually makes computer networks— not the hardware!
  - Protocols: describe *how* two or more communicating parties exchange information.
  - Services: describe *what* a network offers to parties that want to communicate.
  - Interfaces: describe how a client can make use of network services, i.e. how the services can be accessed.

# Internet As an Example



PC



server



wireless laptop



cellular handheld

- *end systems*: millions of connected computing devices

- *communication links*

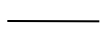
- ❖ fiber, copper, radio, satellite

- *routers*: forward packets (chunks of data)

- *protocols* control sending, receiving of msgs



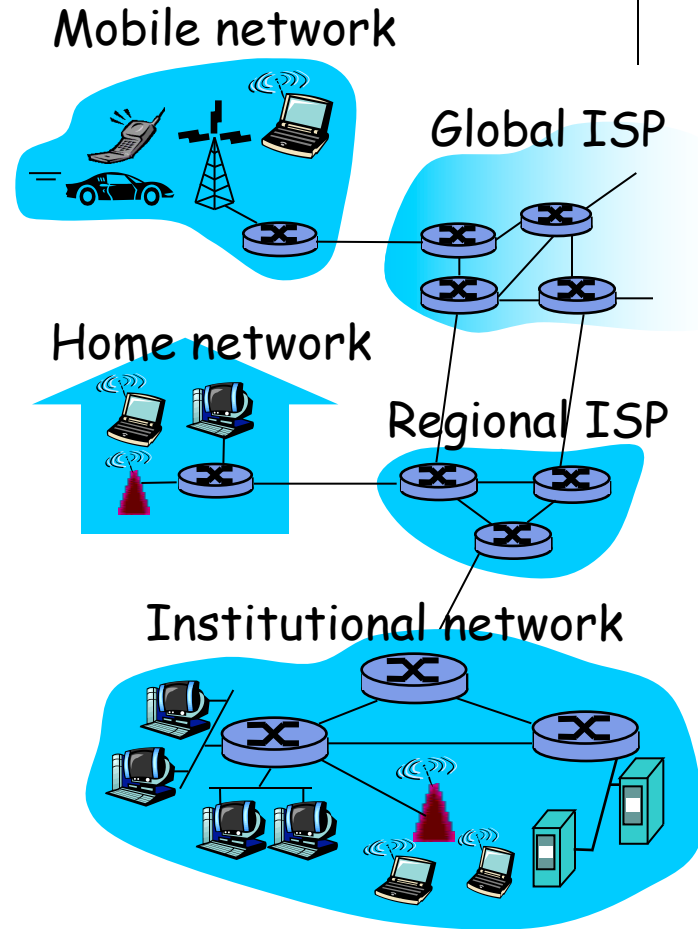
access points

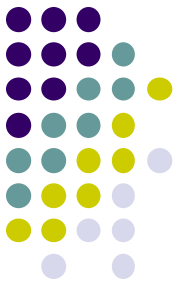


wired links



router

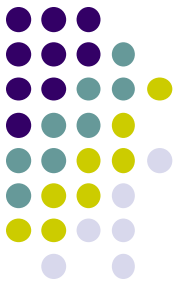




# Chapter 1: roadmap

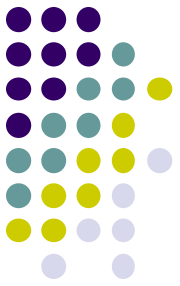
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# The Need for Protocols

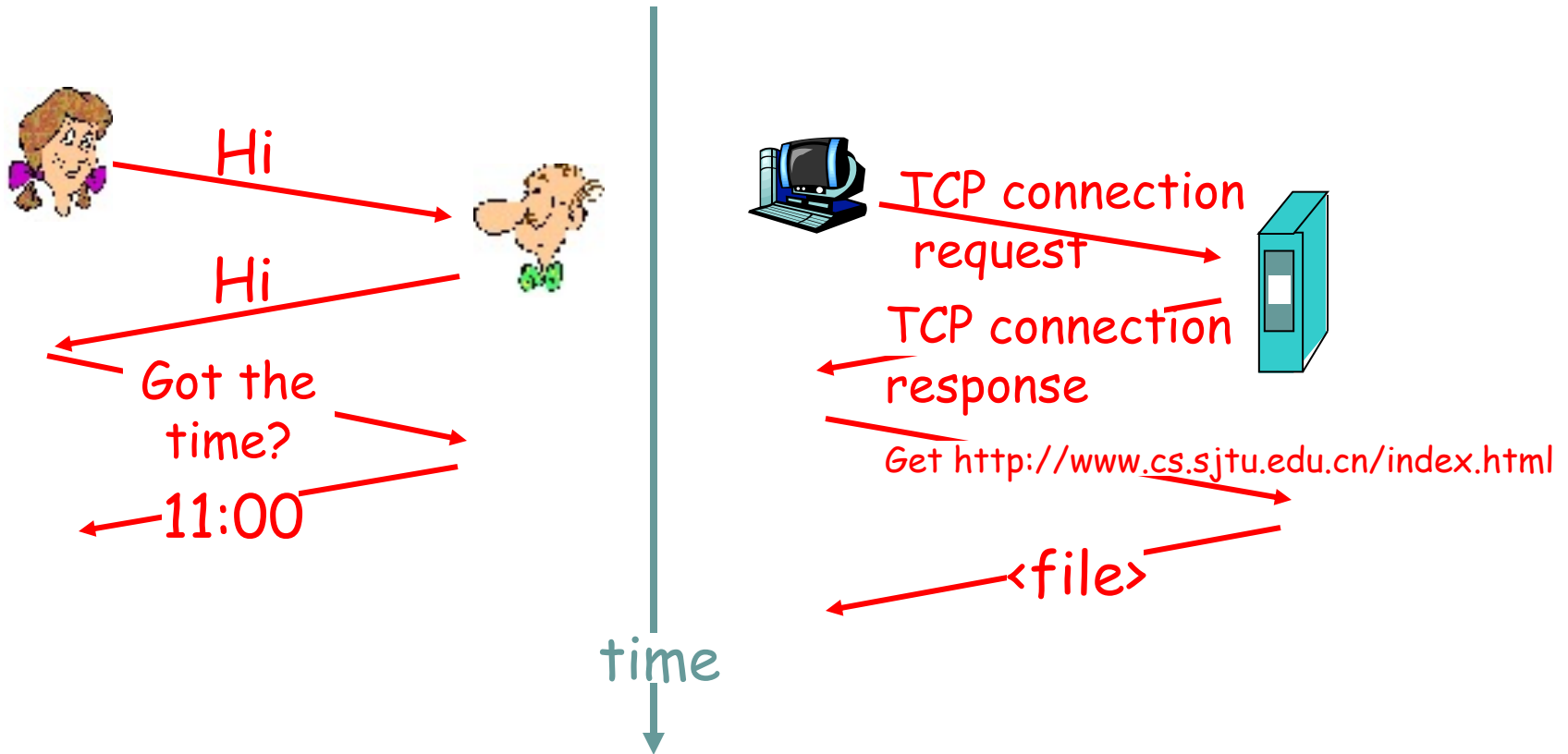


- Basic communication hardware consists of mechanisms that can transfer bits from one point to another. ( **cumbersome and inconvenient** )
- Application programs that use a network don't interact directly with network hardware. Instead, they interact with protocol software that follows the rules of a given protocol.
- A **protocol** defines the format and the order of messages exchanged between two or more communicating entities, as well as the actions taken on the transmission and/or receipt of a message or other event.

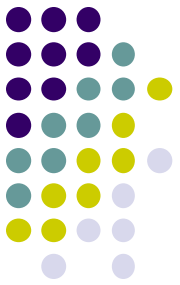
# What's a protocol?



a human protocol and a computer network protocol:



# Protocols are complex



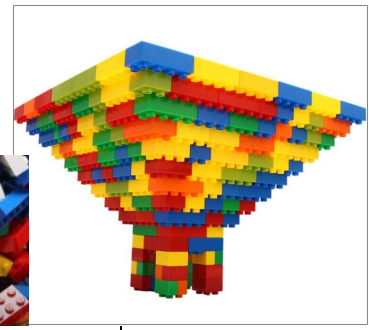
## Networks are complex!

- many “pieces”:
  - hosts
  - routers
  - links of various media
  - applications
  - protocols
  - hardware, software

## Communication are complex!

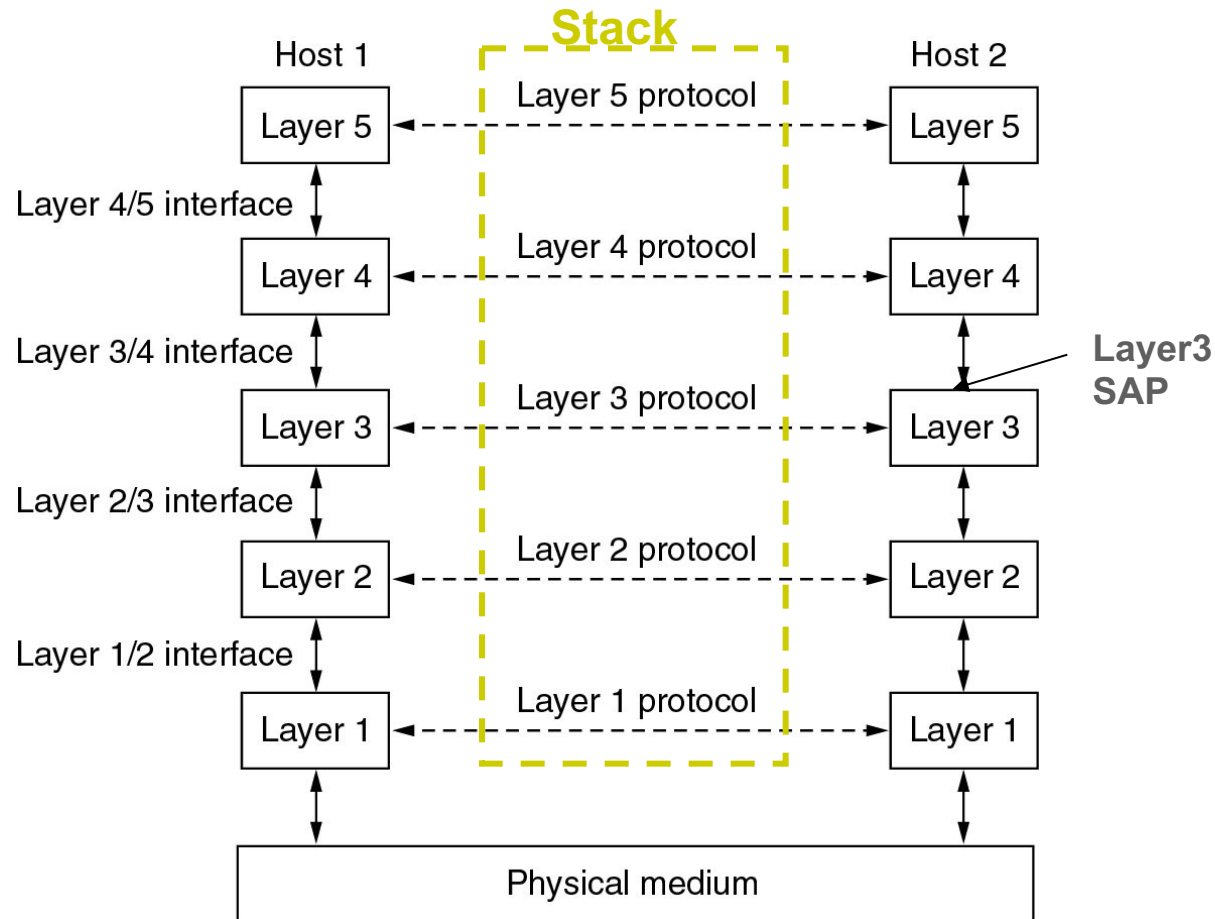
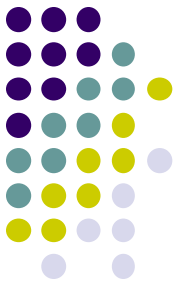
- many “tasks”:
  - data encoding,
  - transportation,
  - addressing,
  - error control,
  - flow control,
  - congestion control,
  - Media Access Control

# Why Protocol Layering



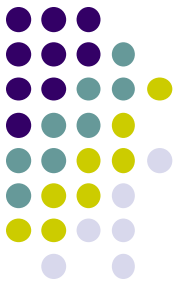
- most network software are organized as a stack of layers or levels, each one built upon the one below it.
  - To **reduce design complexity**, divide the communication problem into subpieces and to design a separate protocol for each subpiece, making each protocol easier to design, analyze and implement.
  - **Independence**. Each layer could be designed, maintained and updated independently, as long as keep in mind the services the lower layer provides for it and the services it should provide for the upper layer.
  - **Flexibility**. Allow subsets of protocols be used as needed and allow any one of the protocols be replaced or updated.

# Layering Model



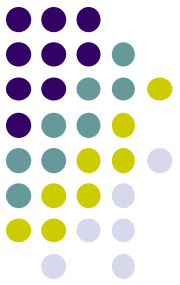
- Layers, peers, protocols, services, interfaces and stack.

# Concepts of Layering



- **Protocol**: two parties at different sites, but at the same level (peers), always agree on how they will exchange information.
- In order for one party to send and receive information, it can only make use of the **communication services** offered by the layer directly underneath it.
- Services offered by a layer are always fully specified in terms of an **interface** that makes those services accessible.

# How Layered Software Works



- Encapsulation: multiple, nested Headers
  - Protocol software in a given layer on the sending computer adds information (header) to the outgoing data, and software in the same layer on the receiving computer uses the header to process incoming data.
  - Outgoing data passes down through each layer, with headers added, and incoming data passes up through each layer, with headers verified and removed.

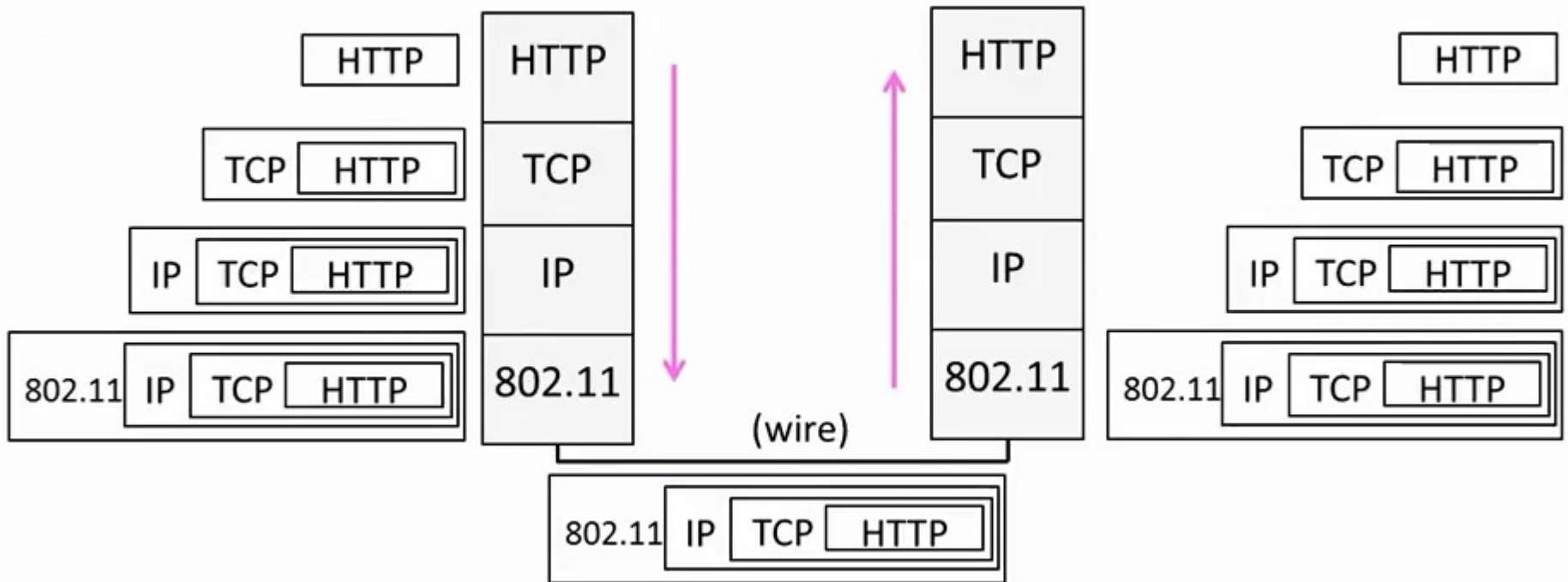
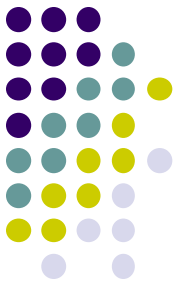
Ethernet Frame

HTTP GET

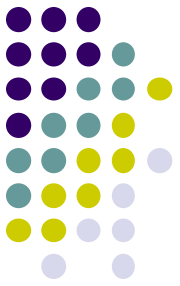
|                                                              |                                                   |                                        |                                              |
|--------------------------------------------------------------|---------------------------------------------------|----------------------------------------|----------------------------------------------|
| Src: 00:e0:81:10:19:fc<br>Dst: 00:a0:cc:54:1d:4e<br>Type: IP | Src: 192.168.0.40<br>Dst: 192.168.0.50<br>TTL: 30 | Src: 1081<br>Dst: 80<br>Chksum: 0xa858 | GET /index.htm HTTP/1.1<br>Host: sjtu.edu.cn |
|--------------------------------------------------------------|---------------------------------------------------|----------------------------------------|----------------------------------------------|

# How Layered Software Works

## Encapsulation

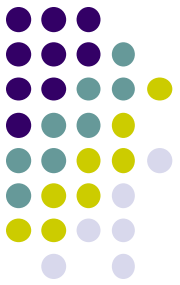


# Services: Connections or Not



- **Connection-oriented:** This is the telephone model: you first **establish** a connection, then do a lot communication, and finally **release** the connection.
- **Connectionless:** The postal model: your data is put into some kind of envelope on which the destination address has been written. The envelope +contents is sent to the destination.

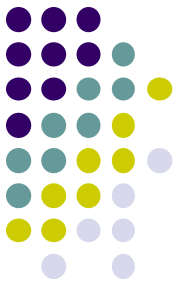
# Services: Reliable or Not



- Reliable : no bit error, no data loss, in order
- Reliable service is implemented by having the receiver acknowledge the receipt of each message. Performance may degrade.

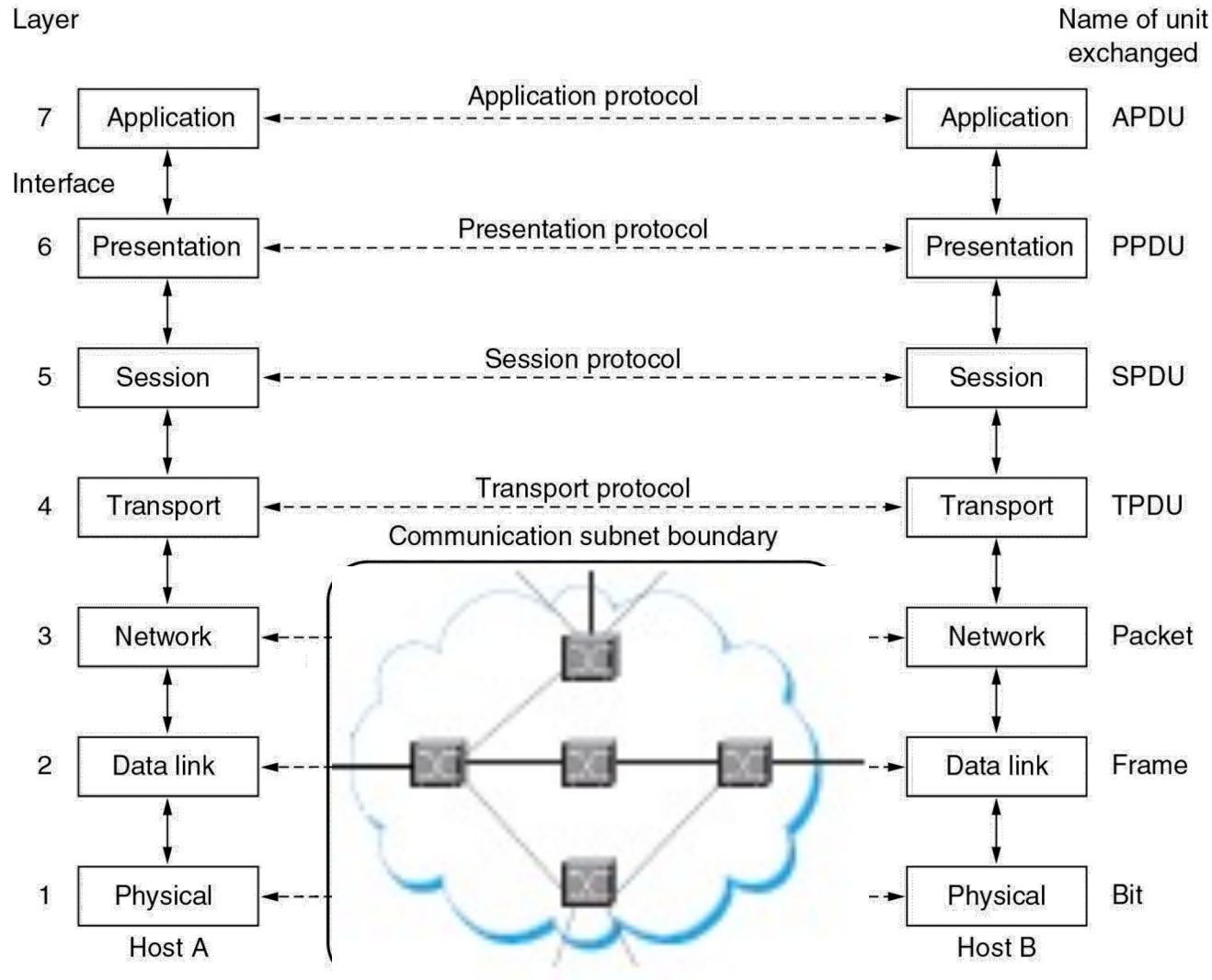
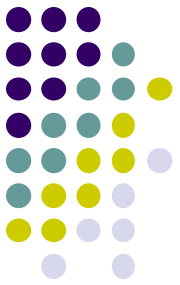
| Service               | Example          | Connection          |
|-----------------------|------------------|---------------------|
| Reliable connection   | TCP (www, email) | Connection-Oriented |
| Unreliable connection | Voice over IP    | Connection-Oriented |
| Reliable datagram     | RUDP             | Connectionless      |
| Unreliable datagram   | IP, UDP (DNS)    | Connectionless      |

# Layering/Service/Reference Models

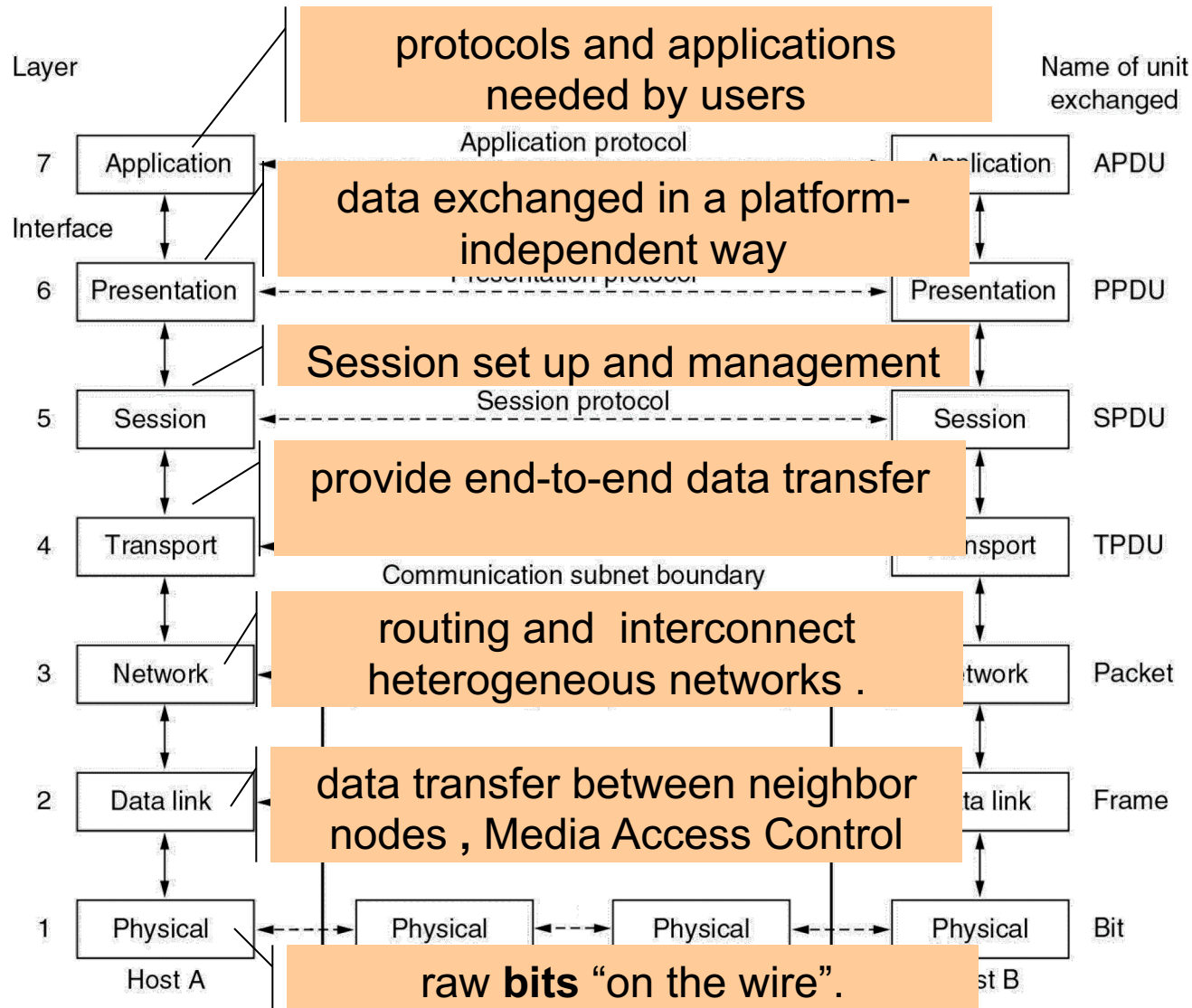
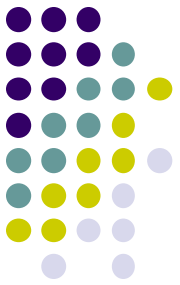


- The OSI Reference Model
- The TCP/IP Reference Model
- A Comparison of OSI and TCP/IP

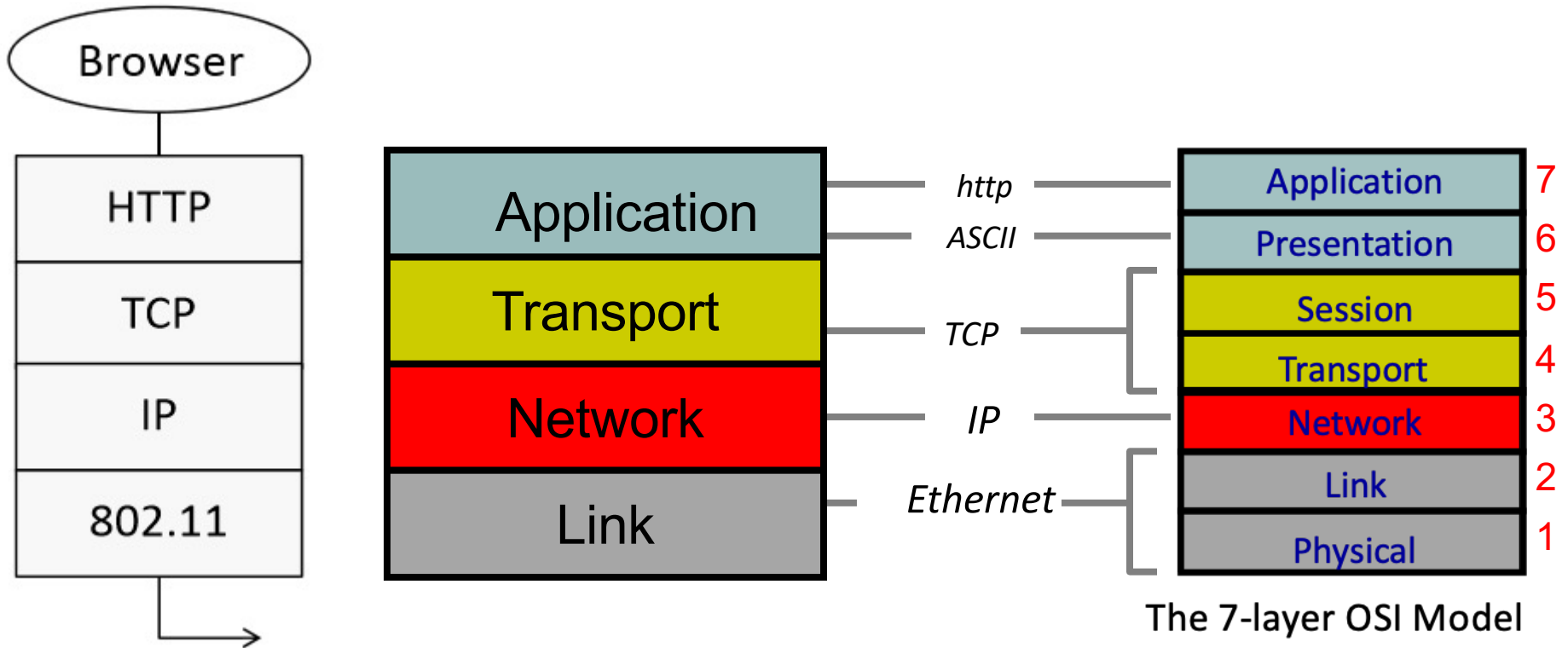
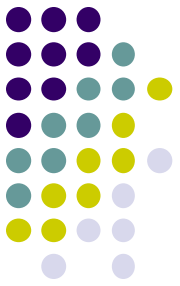
# The OSI 7-layer Reference Model.



# The OSI 7-layer Reference Model.



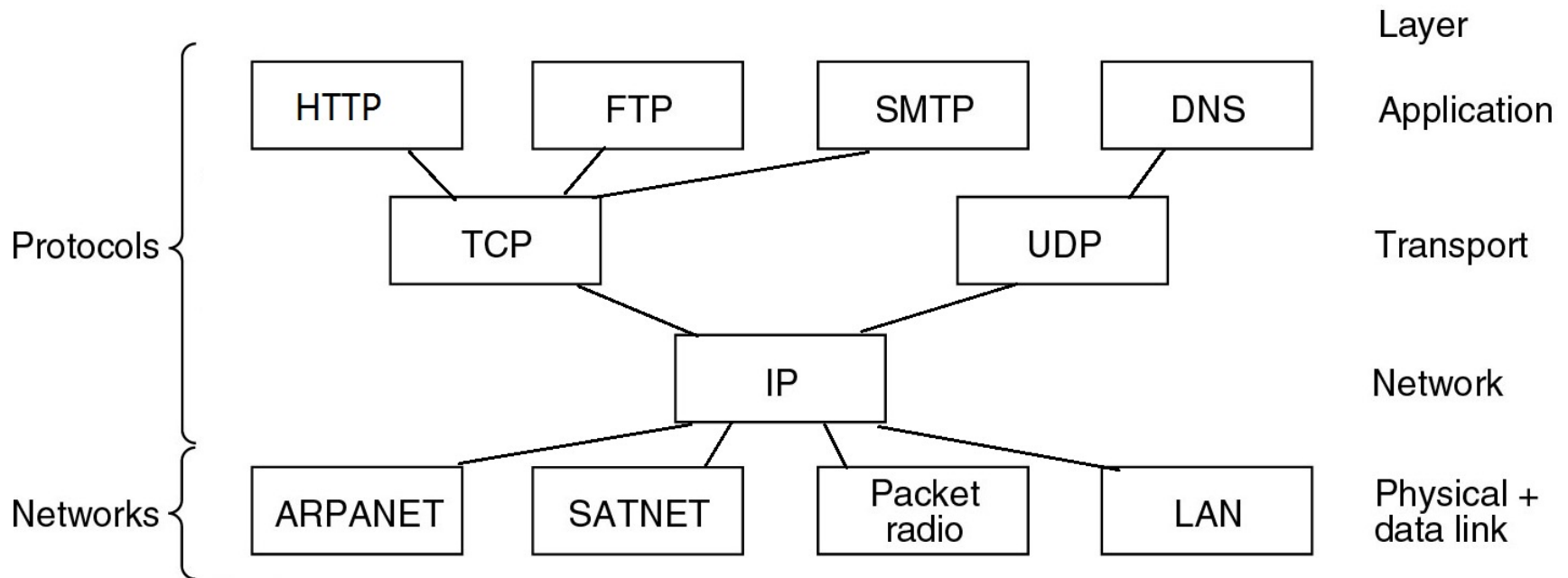
# The 4-layer TCP/IP Model



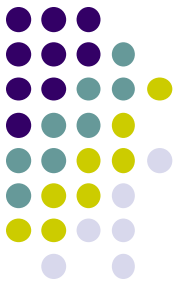
# “Hourglass” philosophy of Internet



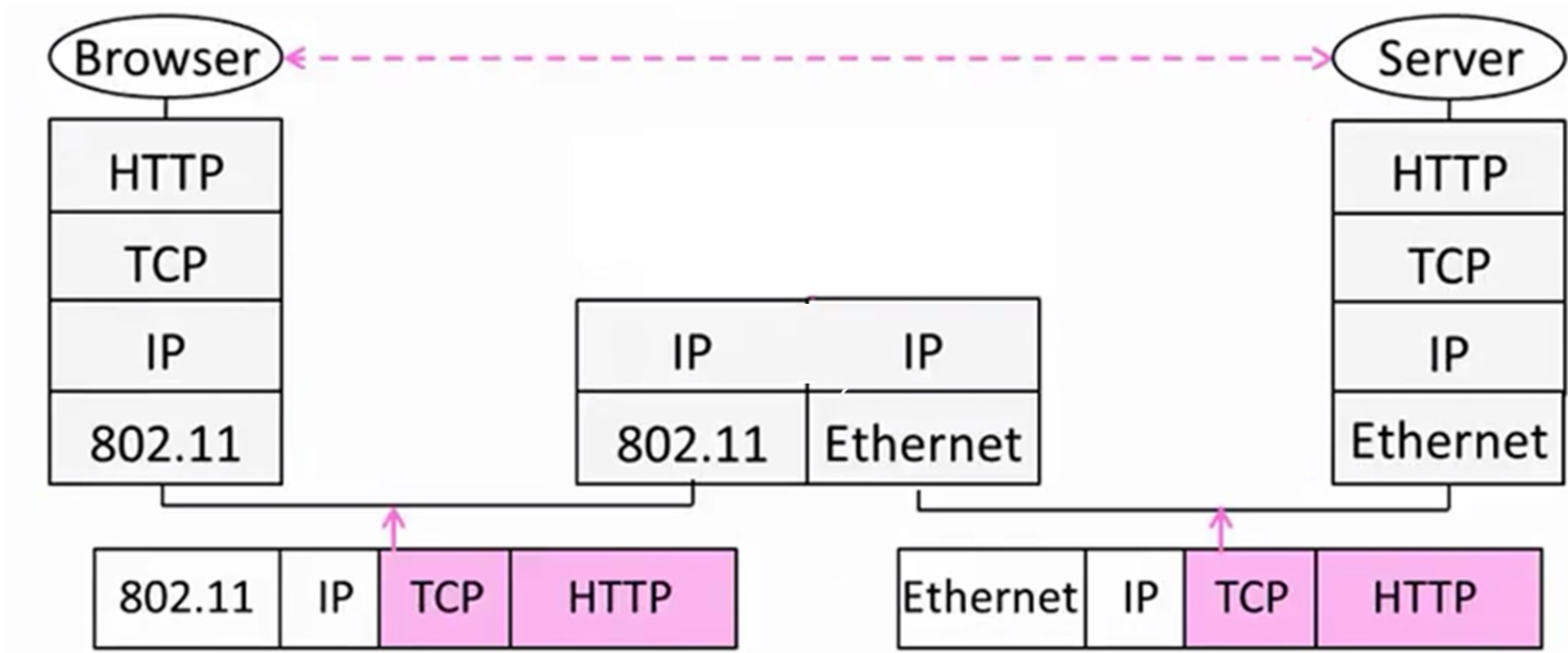
IP bridges different applications over different networks.



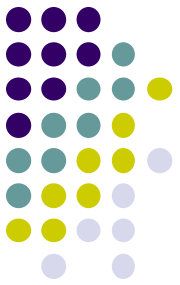
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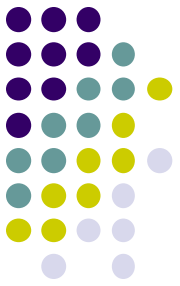


# Comparing OSI and TCP/IP Models



- Much in common:
  - a stack of independent protocols
  - functionality of the layers is roughly similar
- Many differences:
  - Services, interfaces and protocols are central concepts of the OSI model, TCP/IP model doesn't distinguish these concepts and is not a general model.
  - OSI model was devised before the corresponding protocols, with TCP/IP the reverse was true.
  - OSI model/protocols took too much time and are too complex, while TCP/IP is simple and not so comprehensive.
  - Number of layers
- OSI model has proven to be exceptionally useful for discussing computer networks, OSI protocols have not become popular.
- TCP/IP Model is practically nonexistent, but protocols are widely used, deeply entrenched, and thus hard to replace.

# Who takes over the world

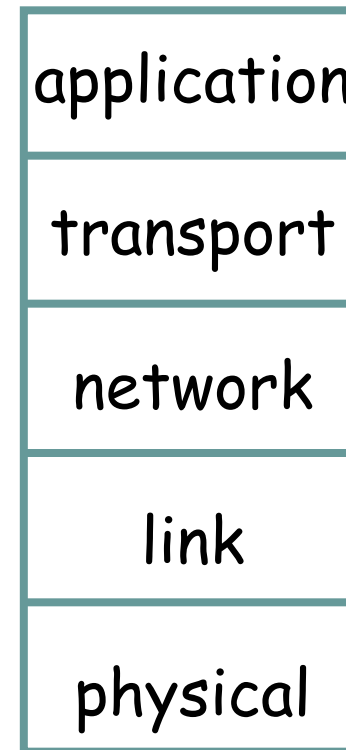


- Why OSI not
  - Bad timing
  - Bad technology
  - Bad implementations
- Why TCP/IP protocol suite is successful
  - it was there when needed
  - freely distributed with the UNIX operating system.

# Hybrid Model Used in this Course



- **application**: programs using network services (**http, ftp, smtp**)
- **transport**: end-end data transfer (**tcp, udp**)
- **network**: send packets over multiple networks (**ip, routing algorithms**)
- **link**: data transfer between neighboring network nodes (**Ethernet, WiFi, ppp, MAC**)
- **physical**: send bits as signals “on the wire” (**media, modulation, encoding**)



Unit Name

Message

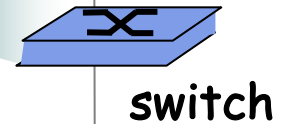
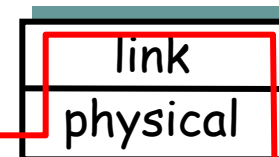
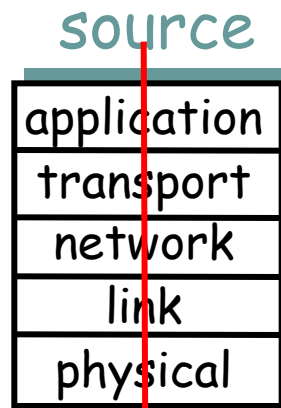
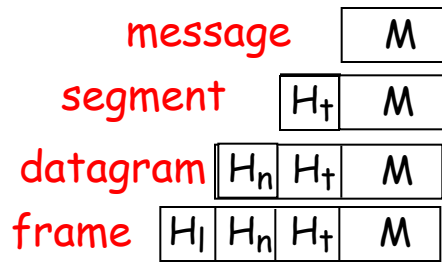
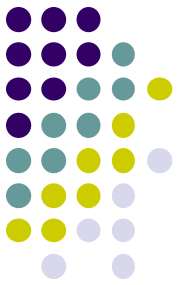
Segment

Datagram  
/Packet

Frame

bit

# Protocol Walkthrough



the Internet architecture puts much of its complexity at the edges of the network

